



A note on the density of zeros of partial sums of the Dirichlet lambda, beta and eta functions

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Abstract

The aim of this note is to provide a different and less technical proof of the density of the real parts of zeros of partial sums of the Dirichlet eta function compared to that presented by Farag. To this end, by leveraging Bohr's equivalence theorem, we establish that the zeros of the partial sums of the Dirichlet lambda and beta functions also exhibit real parts that are dense in the same critical interval as those of the Riemann zeta function. Concurrently, we provide information on the values of the bounds of the critical intervals for the partial sums of the Riemann zeta function and the Dirichlet eta function.

Keywords Dirichlet polynomials · Dirichlet character · Multiplicative functions · Riemann's zeta function · Dirichlet lambda and beta functions · Dirichlet eta function

Mathematics Subject Classification 11M06 · 11M26 · 11L03 · 11R42 · 30C15 · 30Axx

1 Introduction

A *general Dirichlet series* is an infinite series that takes the form

$$\sum_{n=1}^{\infty} a_n e^{-\lambda_n s}, \quad a_n \in \mathbb{C}, \quad (1)$$

where $\{\lambda_n\}$ is a strictly increasing sequence of non-negative numbers tending to infinity. Associated to it (see [2]), we find the half-planes of convergence $\{s \in \mathbb{C} : \operatorname{Re}(s) > \sigma_c\}$ and absolute convergence $\{s \in \mathbb{C} : \operatorname{Re}(s) > \sigma_a\}$, where σ_c and σ_a are called the abscissas of convergence and absolute convergence respectively. Truncating (1) we obtain exponential polynomials called *Dirichlet polynomials*. These exponential poly-

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nomials can be normalized to the form

$$P(s) = 1 + \sum_{j=1}^n m_j e^{-w_j s}, \quad n \in \mathbb{N}, \quad (2)$$

with increasing positive numbers $w_1 < \dots < w_n$ and $m_j \in \mathbb{C} \setminus \{0\}$, $j = 1, \dots, n$. So, the partial sum of a Dirichlet series is a Dirichlet polynomial.

Consider two Dirichlet polynomials of type (2) with the same set of exponents $\Lambda = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$, say $f(s) = 1 + \sum_{m=1}^n a_m e^{-\lambda_m s}$ and $g(s) = 1 + \sum_{m=1}^n b_m e^{-\lambda_m s}$. If $B = \{\beta_1, \beta_2, \dots, \beta_{q_n}\}$, where q_n depends on n , is a basis for Λ then, for each $m = 1, 2, \dots, n$, it is verified

$$\lambda_m = \sum_{k=1}^{q_n} r_{m,k} \beta_k, \quad (3)$$

for some rational numbers $r_{m,k}$.

Definition 1.1 We say that two Dirichlet polynomials

$$f(s) = 1 + \sum_{m=1}^n a_m e^{-\lambda_m s} \quad \text{and} \quad g(s) = 1 + \sum_{m=1}^n b_m e^{-\lambda_m s}$$

of type (2) with the same set of exponents are *Bohr-equivalent*, relative to the basis $\{\beta_1, \beta_2, \dots, \beta_{q_n}\}$, if for some set of real numbers $Y = \{y_1, y_2, \dots, y_{q_n}\}$ it is satisfied

$$b_m = a_m \exp\left(i \sum_{k=1}^{q_n} r_{m,k} y_k\right), \quad \text{for each } m = 1, 2, \dots, n,$$

where the $r_{m,k}$'s are given by (3).

Lemma 1.2 ([8, Proposition 2.2]) *Two ordinary Dirichlet polynomials*

$$P(s) = 1 + \sum_{m=2}^n a_m m^{-s} \quad \text{and} \quad Q(s) = 1 + \sum_{m=2}^n b_m m^{-s},$$

are Bohr-equivalent if and only if there exists a function $\chi: \{1, 2, \dots, n\} \rightarrow \mathbb{C}$ such that

- (i) $\chi(1) = 1$ and $\chi(ml) = \chi(m)\chi(l)$ for any positive integers m and l so that $ml \leq n$;
- (ii) $b_m = \chi(m)a_m$ for each $m = 2, \dots, n$;
- (iii) $|\chi(p)| = 1$ whenever $a_m \neq 0$ and p a prime divisor of m .

Given an ordinary Dirichlet polynomial $P(s) = 1 + \sum_{m=2}^n a_m m^{-s}$, it is known that all zeros of $P(s)$ are located on a vertical strip, called the critical strip of $P(s)$, bounded by the real numbers α_P and β_P , where $\alpha_P := \inf\{\operatorname{Re}(s) : P(s) = 0\}$ and $\beta_P := \sup\{\operatorname{Re}(s) : P(s) = 0\}$. These bounds allow us to define an interval $I_P := [\alpha_P, \beta_P]$, called the critical interval of $P(s)$.

Finally, associated with $P(s)$, we will take the set

$$R_P := \overline{\{\operatorname{Re}(s) : P(s) = 0\}},$$

on which we are going to focus our attention from now on.

Lemma 1.3 ([8, Theorem 3.1]) *Let*

$$P(s) = 1 + \sum_{m=2}^n a_m m^{-s} \quad \text{and} \quad Q(s) = 1 + \sum_{m=2}^n b_m m^{-s}$$

be two Bohr-equivalent ordinary Dirichlet polynomials, then $R_P = R_Q$.

As a direct consequence of Lemma 1.3, it is important to remark that two Bohr-equivalent ordinary Dirichlet polynomials, $P(s)$ and $Q(s)$, have the same critical strip. More so, for each zero $s_0 = \sigma_0 + it_0$ of $P(s)$ and for all $\epsilon > 0$, we can find a zero $s'_0 = \sigma'_0 + it'_0$ of $Q(s)$ such that $\sigma_0 - \epsilon < \sigma'_0 < \sigma_0 + \epsilon$.

A new terminology to define this condition on a concrete substrip contained in the critical strip of an ordinary Dirichlet polynomial was introduced in [8].

Definition 1.4 Let $S_{(a,b)} \equiv \{s \in \mathbb{C} : a < \operatorname{Re}(s) < b\}$ be an open vertical strip contained in the critical strip of a Dirichlet polynomial $P(s)$. We say that $P(s)$ has the *density property on the strip $S_{(a,b)}$* (or the $\mathcal{D}_{S_{(a,b)}}$ property) when it has zeros arbitrarily near any vertical line inside the strip $S_{(a,b)}$. That is, given σ_2 with $a < \sigma_2 < b$ and $\epsilon > 0$, a complex number $s^* = \sigma^* + it^*$ can be found such that $\sigma_2 - \epsilon < \sigma^* < \sigma_2 + \epsilon$ and $P(s^*) = 0$.

Equivalently, note that if $P(s)$ is a Dirichlet polynomial which has the density property on the strip $S_{(a,b)}$, this means that the set $\{\operatorname{Re}(s) : P(s) = 0\}$ is dense in the interval $[a, b]$, that is, $R_P = [a, b]$. Hence we have the following result.

Lemma 1.5 ([8, Corollary 3.4]) *Let $S_{(a,b)}$ be an open vertical strip where the ordinary Dirichlet polynomial $P(s)$ satisfies the $\mathcal{D}_{S_{(a,b)}}$ property. If $Q(s)$ is an ordinary Dirichlet polynomial such that $P(s)$ and $Q(s)$ are Bohr-equivalent, then $Q(s)$ has the $\mathcal{D}_{S_{(a,b)}}$ property.*

The study of the zeros of the partial sums of Dirichlet series still represents an important subject of research (see, for example, [11, 12]) and has been generalized to L-functions and the Dedekind zeta function (see [9, 15–17, 21, 22, 24, 25]).

Among the numerous authors who worked with the partial sums of the Riemann zeta function

$$\zeta_n(s) = \sum_{k=1}^n \frac{1}{k^s}, \quad s \in \mathbb{C},$$

we can cite Spira [27–30] and Turán in [31]. Spira also used Bohr's equivalence theorem (see [2, Theorem 8.16, p. 178]) as a tool to study the distribution of zeros of $\zeta_n(s)$ and exposed, in [28], as a conjecture, the possibility that the real parts of the

zeros of $\zeta_n(s)$ are dense in certain intervals included in their respective critical strips. A partial answer to this conjecture was published by Moreno in [23]. He tackled the theme of density by saying that one can find zeros of these partial sums, in the critical strip, close to any line perpendicular to the $\text{Re}(s)$ -axis. This result was generalized in [7] for $\zeta_n(s)$ in the case where n is prime and for any n in [19].

We consider, here, the Dirichlet beta function denoted by $\beta(s)$ and defined by

$$\beta(s) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^s} \quad \text{with } \text{Re}(s) > 0.$$

It admits particular values like $\beta(1) = \pi/4$ associated with $\tan^{-1}(1)$, $\beta(2)$ known as Catalan's constant, or even $\beta(3) = \pi^3/32$.

The Dirichlet beta function has no trivial zeros in its critical strip. We do not know if it admits an infinity of zeros on the critical line nor if all its zeros are on it.

The Dirichlet lambda function is defined by

$$\lambda(s) = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^s} = (1-2^{-s})\zeta(s) \quad \text{with } \text{Re}(s) > 1.$$

Knowledge regarding the values and zeros of $\beta(s)$ and $\lambda(s)$ remains limited, and this area continues to be a subject of active research (see, for example, [1, 6, 13, 14]).

The purpose of this article is threefold. Our initial focus is on the distribution of zeros of the partial sums $\beta_n(s)$ and $\lambda_n(s)$ of $\beta(s)$ and $\lambda(s)$, respectively, employing Bohr's equivalence theorem as a fundamental tool. Secondly, we derive improved estimates for the bounds of the critical interval of the zeros of the partial sums of the Riemann zeta function. Finally, thanks to the equality $2\lambda(s) = \zeta(s) + \eta(s)$, where $\eta(s)$ is the Dirichlet eta function defined by

$$\eta(s) = (1-2^{1-s})\zeta(s) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^s} \quad \text{with } \text{Re}(s) > 0,$$

we establish the density of the real parts of zeros for partial sums of the Dirichlet eta function and recover Farag's result [10] as a corollary. We also improve the values of the bounds of the critical interval of such truncated Dirichlet series

$$\eta_n(s) = \sum_{k=1}^n \frac{(-1)^{k-1}}{k^s} \quad \text{with } \text{Re}(s) > 0.$$

2 Main results

Definition 2.1 Let q be a positive integer. We define a *Dirichlet character modulo q* as a function $\chi_q: \mathbb{Z} \rightarrow \mathbb{C}$ which is periodic of period q , i.e. $\chi_q(n+q) = \chi_q(n)$ for all n in $\mathbb{Z}$, completely multiplicative, i.e. $\chi_q(nm) = \chi_q(n)\chi_q(m)$ for all n, m in $\mathbb{Z}$, and

which also satisfies $\chi_q(1) = 1$ and $\chi_q(n) = 0$ when $(n, q) > 1$, where (n, q) is the greatest common divisor of n and q .

Let ϕ be the totient function (for a definition see [3, Paragraph 2.3, Equation (1), p. 25]). If $q = 4$, then $\phi(4) = 2$ which means that there are two Dirichlet characters modulo 4 that we write χ_4^0 and χ_4^1 such that

$$\chi_4^0(n) = \begin{cases} 1 & \text{if } (n, 4) = 1, \\ 0 & \text{if } (n, 4) > 1, \end{cases}$$

is the trivial character and

$$\chi_4^1(n) = \begin{cases} 1 & \text{if } n \equiv 1(4), \\ -1 & \text{if } n \equiv 3(4), \\ 0 & \text{if } 2 \mid n. \end{cases}$$

Let $a_{\zeta_n} = \inf \{\operatorname{Re}(s) : \zeta_n(s) = 0\}$ and $b_{\zeta_n} = \sup \{\operatorname{Re}(s) : \zeta_n(s) = 0\}$ then $[a_{\zeta_n}, b_{\zeta_n}]$ is the critical interval of the partial sums of the Riemann zeta function.

Theorem 2.2 *The partial sums of the Dirichlet lambda and beta functions have all their zeros in the critical strip of ζ_n and their real parts are dense in the critical interval $[a_{\zeta_n}, b_{\zeta_n}]$.*

Proof The Dirichlet beta and lambda functions are L-functions associated respectively with the characters χ_4^1 and χ_4^0 . Indeed, we have

$$\beta(s) = \sum_{n=1}^{\infty} \frac{\chi_4^1(n)}{n^s} = \prod_{n=2}^{\infty} \left(1 - \frac{\chi_4^1(p_n)}{p_n^s}\right)^{-1}$$

for $\operatorname{Re}(s) \geq 1$ where the p_n are prime numbers and

$$\lambda(s) = \sum_{n=1}^{\infty} \frac{\chi_4^0(n)}{n^s},$$

where the Dirichlet characters χ_4^0 and χ_4^1 are completely multiplicative functions.

Let

$$\lambda_n(s) = \sum_{k=1}^n \frac{\chi_4^0(k)}{k^s} \quad \text{and} \quad \beta_n(s) = \sum_{k=1}^n \frac{\chi_4^1(k)}{k^s}$$

be the partial sums of the Dirichlet lambda and beta functions, respectively. We can say, thanks to Lemma 1.2, that they are equivalent in the sense of Bohr, as Dirichlet polynomials, to the partial sums of the Riemann zeta function $\zeta_n(s)$.

We know (see [7, 19]) that for any $n \geq 2$, $\zeta_n(s)$ has the density property, as we defined it previously in Definition 1.4, in the critical strip $\mathcal{S}_{(a_{\zeta_n}, b_{\zeta_n})}$ where

$$a_{\zeta_n} = -\frac{\log 2}{\log\left(\frac{n-1}{n-2}\right)} + \Delta_n \quad \text{for } n > 2 \quad \text{with } \limsup_{n \rightarrow \infty} |\Delta_n| \leq \log 2$$

(see [20]) and

$$b_{\zeta_n} = 1 + \left(\frac{4}{\pi} - 1 + o(1)\right) \frac{\log \log n}{\log n}, \quad n \rightarrow \infty$$

(see [18]). In conclusion, using Lemmas 1.3 and 1.5, the theorem holds. $\square$

In the article [7, Proposition 1], it was proved that if $n+1$ is a prime number greater than 2 then

$$R_{\zeta_n} \cap \left[a_{\zeta_n}, \frac{\log 2}{\log\left(1 + \frac{1}{n}\right)} \right] \subset R_{\zeta_{n+1}}.$$

Using the previous theorem, we can provide additional information about the critical interval of $\zeta_n(s)$.

Corollary 2.3 *Let $n > 2$ be an odd integer. Then we have*

$$[a_{\zeta_n}, b_{\zeta_n}] = [a_{\zeta_{n+1}}, b_{\zeta_{n+1}}].$$

Proof By definition of the partial sums of the Dirichlet lambda function $\lambda_n(s)$, we can say that if $n > 2$ is an odd integer then we have $\lambda_n(s) = \lambda_{n+1}(s)$ for all $s \in \mathbb{C}$ which implies that they have the same zeros and therefore $R_{\lambda_n} = R_{\lambda_{n+1}}$. By Theorem 2.2 we can say that λ_n is Bohr equivalent to ζ_n for all $n \geq 2$. Thus, by Lemma 1.3, we can conclude that $R_{\lambda_n} = R_{\zeta_n}$. Moreover, we know (see [7, 19]) that ζ_n has the density property which allows us to write that $R_{\zeta_n} = [a_{\zeta_n}, b_{\zeta_n}]$ and the result holds. $\square$

Having addressed the first two points, we conclude by examining the third and final aspect of this note. Our objective is to extend the density results for the real parts of zeros of partial sums from the Riemann zeta or Dirichlet lambda functions to the partial sums of the Dirichlet eta function. However, the unique characteristics of the Dirichlet eta function, including its non-L-function status and then the inapplicability of Bohr's equivalence theorem, necessitate novel approaches.

We denote by $[a_{\eta_n}, b_{\eta_n}]$ the critical interval of the partial sums of the Dirichlet eta function $\eta(s)$. For $n = 3$, Borwein, Fee, Ferguson and Van der Waal in [5, Theorem 4.3, p.28] proved that the real parts of the zeros of $\zeta_3(s)$ are dense in the interval $[a_{\zeta_3}, b_{\zeta_3}] = [-1, r_1]$ where r_1 is the real solution of the equation $1 - 2^{-x} - 3^{-x} = 0$. It turns out that Mora, in [21, p.5], proved that $R_{\eta_3} = [-1, b_1]$ where b_1 is the real solution of the equation $2^{-c} + 3^{-c} = 1$. Now, for $n = 3$, we know, thanks to [8, Proposition 4.5], that $\eta_3(s)$ has the density property in its critical strip, i.e. $R_{\eta_3} = [a_{\eta_3}, b_{\eta_3}]$. So, finally, $[a_{\eta_3}, b_{\eta_3}] = [a_{\zeta_3}, b_{\zeta_3}]$ and this first partial result seems to be going in the right direction.

The theorem we are about to state yields the desired result. While we do not obtain density throughout the entire critical interval of $\eta_n(s)$, we do improve upon the known bounds of this interval.

Theorem 2.4 *Let $n > 2$ be an integer. The real parts of the zeros of the partial sums of the Dirichlet eta function are dense in the interval $[a_{\zeta_n}, b_{\zeta_n}]$.*

Proof We proved that the Dirichlet series $\lambda(s)$ and $\zeta(s)$ are Bohr equivalent and are absolutely convergent for $\sigma > 1$. They take, thus, the same values in each vertical strip $\mathcal{S}_{(a,b)}$ included in their domain of absolute convergence. More precisely, $\lambda_n(s)$ takes the same values as $\zeta_n(s)$ in any half-plane $\operatorname{Re}(s) < b$. It is also interesting to notice that these three functions $\lambda_n(s)$, $\zeta_n(s)$ and $\eta_n(s)$ as well as their derived functions are, as almost periodic functions, bounded in any vertical strip included in their critical strip $\mathcal{S}_{(a,b)}$.

We use, now, the following equality

$$2\lambda_n(s) = \zeta_n(s) + \eta_n(s), \quad \text{for all } s \in \mathbb{C} \text{ and } n > 2. \quad (4)$$

The zero sets of $2\lambda_n(s)$ and $\lambda_n(s)$ are the same so, by Theorem 2.2, for all $n > 2$, the function $2\lambda_n(s)$ has the real parts of its zeros dense in its critical interval which turns out to be the same critical interval of $\lambda_n(s)$ and $\zeta_n(s)$. So we can say that $2\lambda_n(s)$ and $\zeta_n(s)$ have a zero on each vertical line inside their critical strip.

Using reductio ad absurdum, suppose, now, that for all $n > 2$, η_n has no zero in the vertical strip $[a_{\zeta_n}, b_{\zeta_n}] \times \mathbb{R}$. Let $s_0 \in [a_{\zeta_n}, b_{\zeta_n}] \times \mathbb{R}$ be a zero of ζ_n with $\operatorname{Re}(s_0) = \sigma_0$ such that it is not a zero of $2\lambda_n$. By [26, Theorem 3.1, p. 493] the fact that σ_0 belongs to R_{ζ_n} assures us of the existence of a sequence $\{t_j\}_{j=1,\dots}$ of real numbers such as $\lim_{j \rightarrow \infty} \zeta_n(\sigma_0 + it_j) = 0$. Using equality (4), we get that

$$|2\lambda_n(\sigma_0 + it_j) - \eta_n(\sigma_0 + it_j)| < \epsilon < |\eta_n(\sigma_0 + it_j)|.$$

By Rouché's theorem (see [4, p. 88]), η_n and $2\lambda_n$ therefore have the same number of zeros on this vertical line. As $R_{\zeta_n} = R_{2\lambda_n}$ then by density, $2\lambda_n$ has a zero on each vertical line included in its critical band. Thus, we obtain a contradiction with the fact that η_n has no zero in the vertical band $[a_{\zeta_n}, b_{\zeta_n}] \times \mathbb{R}$ and the result holds. $\square$

Then, we can give, as immediate consequences of Theorem 2.4 and of [21, Corollary 4.1 and Theorem 4.1], new relations between the bounds of the critical intervals of the partial sums of the Riemann zeta function $\zeta(s)$ and of the Dirichlet eta function $\eta(s)$.

Corollary 2.5 *Let $n > 2$ be an integer. We have*

$$a_{\eta_n} \leq a_{\zeta_n} < 0 \quad \text{and} \quad b_{\zeta_n} \leq b_{\eta_n}.$$

Thus, we obtain an immediate information on the density interval of zeros of $\eta_n(s)$.

Corollary 2.6 *For any integer $n > 2$, $[a_{\zeta_n}, b_{\eta_n}]$ is a density interval of zeros of $\eta_n(s)$.*

Corollary 2.6 substantially enhances the left bound of the density interval for $\eta_n(s)$ compared to the result in [21, Theorem 4.1]. Specifically, it provides a concrete value for the previously mentioned $\alpha_n < 0$.

Finally, in [10, Theorem 1], Farag proved that the partial sums of the Dirichlet eta function have their zeros close to any vertical line in the vertical strip $(\sigma, 1) \times \mathbb{R}$ where $\sigma \in (0, 1)$. In other words, the partial sums $\eta_n(s)$ have the density property in $(\sigma, 1) \times \mathbb{R}$. We can, now, write his main theorem ([10, Theorem 1]) as a corollary of our previous result. We propose, thus, a different proof from Farag's original article but also less technical than that given by Mora (see [21] where analytic varieties $|\eta_n^*(z)| = p_n^{-c}$ are introduced with $n > 2$, $c \in \mathbb{R}$ and $\eta_n^*(z) = \eta_n(z) - \frac{(-1)^{p_n-1}}{p_n^z}$).

Corollary 2.7 *We consider the function $\eta_n(s) = \sum_{k=1}^n \frac{(-1)^{k-1}}{k^s}$ for all $n > 2$. Let $\sigma \in (0, 1)$ be a real number. Then there exists N_σ depending only on the interval $(\sigma, 1)$, such that, for all $n > N_\sigma$, $\eta_n(s)$ has an infinity of zeros $s'_n = \sigma'_n + it'_n$ in the critical strip. In particular, given $\tilde{\sigma} \in (\sigma, 1)$, $\epsilon > 0$, there exists a zero $\sigma'_n + it'_n$ such that $|\tilde{\sigma} - \sigma'_n| < \epsilon$.*

Proof Using [18, 20], we have, for all $n > 2$, $a_{\zeta_n} < 0$, $b_{\zeta_n} > 0$ and by Theorem 2.4 each $\eta_n(s)$ has an infinity of zeros in the critical strip $(0, 1) \times \mathbb{R}$. The second part of the corollary is an immediate consequence of Hurwitz's theorem as we can find it in [31, p. 9]. $\square$

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